Agile S/W Process Model

SCRUM

Scrum

- Not an acronym – derived from observing a rugby scrum
- Iterative software development process
  - Empirically controlled (rather than rigid, plan-based)
  - Focuses on maximizing business value
  - Constant improvement and priority/risk driven

Scrum is an empirical process

- Empirical processes do not specify software development activities in advance….
  - Treat the development process as a black box.
  - Development starts with a *best estimate at what will work*. Then the process is regularly inspected at defined intervals (i.e., every day), assessed, and adjusted, if necessary.
  - Process is not well-defined, but based around techniques that focus on *learning from past experience*.
  - Knowledge is used for *guidance going forward*, not for repeatability of past actions.
  - Requires a high level of trust from the client

Roles in Scrum

- Product Owner
  - Manages the vision, ROI and decides on when to release the product into a production environment,
  - Understands business level risks as well as consequences of actions/ decisions from a business perspective,
  - Develops and maintains the product backlog – team assists.
- *Scrum Master*
  - Owns the process, motivates/coaches/guides team,
  - Ensures that the team does not go off target -- makes all project level decisions and ensures that the team has the necessary environment, information to deliver on the vision within the constraints set.
- Scrum team member
  - Plan tasks, manages their commitments and works as a team to achieve the vision set for them. Responsible for delivery of functionality.

Product Backlog

- Product backlog
  - Features, architecture, defects, quality attributes, constraints, etc.
  - Everything currently assessed to be necessary to bring project to completion (could be features not yet implemented, bugs not yet resolved, documents not yet written, etc.)
  - Owned by the 'product manager or product owner' – single role ownership
  - Non-binding (effort) estimates for each item in the backlog are put forward by the product owner/manager – the team assists in this effort.

Scrum Team and Sprinting

- Once the product backlog is ready and estimates put forward - the team starts the sprint (i.e. the iteration), during which they will address chosen backlog items (the sprint backlog)
  - Sports metaphor – once the game starts, the rules cannot be changed, i.e. the sprint (product) backlog cannot be revised (until end of the sprint).
  - Each sprint is *30 calendar working days* (approx. 1½ months in business working days).
  - The team is *self-organising* (as is a footy team) – under the guidance of the *scrum master* (this is a role – does not mean a separate person is needed 100% for this).
  - Team expands backlog into tasks and gets to work on them.
  - Team should be 7 members (+/- 2) [less than 5 discouraged]
    - Hypothesis – *odd member teams work better*

Sprint – Before and After

- Before each sprint session – the team takes time to plan for the sprint
  - Product owner, Scrum master and team meet to prioritise the functional requirements and organise the work,
  - The team identifies the set of functionality they can deliver – the choice is made by assessing business value of the functionality; this is the **sprint backlog**
  - Ends in a short (but formal) presentation outlining the commitment from the team – any stakeholder may attend this session.
- After the sprint – the team reviews the previous iteration (sprint) and identifies what they can learn from it all
  - Similar to an iteration retrospective but includes a presentation of the work done in the previous iteration (i.e. a product functionality demonstration)

Sprint Planning

- The following activities are undertaken in a sprint planning session
  - ☐ Sprint planning meeting time/date set (Scrum master)
  - ☐ Product backlog is presented, with any information helpful to the team from the business (Product owner)
  - ☐ Select product backlog items for the sprint (Entire Team)
  - ☐ Define the goal (vision) for the sprint (Product Owner)
  - ☐ Construct sprint backlog (Scrum team + Scrum Master)
  - ☐ Formal presentation to stakeholders (Scrum master)

Daily Scrum

- In short it is nothing more than a *stand-up meeting*
- Every team member answers the following questions:
  1. What did you do since the last Scrum?
  2. What are you doing until the next Scrum?
  3. What prevented you from doing work?
- No detailed discussions in this forum – the aim here is to *increase transparency*, not debate!
- Must be held at the 'same time, same location' every day
- Scrum master responsible for correcting information gained from Question 3
- Some quick decisions may be taken, left to the discretion of the scrum master (works best just for information gathering)

Managing Sprints

- Burn-down charts are used to get an idea of progress – quantitative data is essential for this
  - Burn-down charts use estimated hours remaining on the project (in terms of work/effort)
  - The slope of the line is used to derive *team velocity*
- Information gained in daily meeting used to cross-verify data being shown on burn-down chart
- A number of sprints is generally undertaken before a product is released

Challenges when using Scrum

- Need a good, effective manager
  - Must be willing to make positive changes (quickly)
  - Must constantly monitor both quantitatively and qualitatively
- Team makes most decisions normally made by managers in traditional approaches – challenging to implement in practice
  - Managers are not generally encouraged to lead this way
- Not every person is happy to accept responsibility for their own actions, the amount of transparency makes it hard to be average
  - Some people work better in a structured environment where work is delegated – not everyone is comfortable making decisions

Benefits of Scrum

- A *transparent development process*
- Productivity increases – once transition is made
- Improvements are slow, but continuous
  - The transparency forces the team to take corrective actions – a sports team would not intentionally take actions to lose
- Accepts software development is not always a predictable, stable and repeatable process
  - Progress is made even when requirements are evolving
  - Constant feedback, empirical evidence to guide team
  - Team communicates better

User Story

- In Scrum projects, the Product Backlog is a list of user stories.
- These User Stories are prioritized and taken into the Sprint Backlog in the Sprint Planning Meeting.
- User Story is narrated from user perspective regarding what he wants to have rather that what system can do for him.

The User Story Structure

As a <*Type of User*>,

I want <*To Perform Some Task*>,

So that <*I can achieve some goal/benefit/value.*>

User Story: Import Gmail Contacts

As a registered user,

I want to automatically follow all of my Gmail contacts that have twitter accounts,

So that I can easily be able to follow my friends' tweets.

From User Story to Tasks

- ❑ Add a new **Import Contacts** page
- ❑ Add a new **Gmail authentication** screen
- ❑ Add a service to **get contacts from Gmail**
- ❑ Add a **Twitter authentication** screen
- ❑ Modify the **contact** class to accommodate Gmail specific data
- ❑ Modify the **contacts data schema**
- ❑ **Save contacts** to the database

Roles

Product Owner: Set priorities

ScrumMaster: Manage process, re--move blocks

Team: Develop product

Stakeholders: observe & advise

Key Artifacts

Product Backlog
- List of requirements & issues
- Owned by Product Owner
- Anybody can add to it
- Only Product Owner prioritizes

Sprint Backlog
- List of tasks
- Owned by team
- Only team modifies it

Burn Down Chart
- Shows estimated vs actual effort of the scrum tasks
- Shows the work remaining

Sprint Goal
- One-sentence summary
- Declared by Product Owner
- Accepted by team

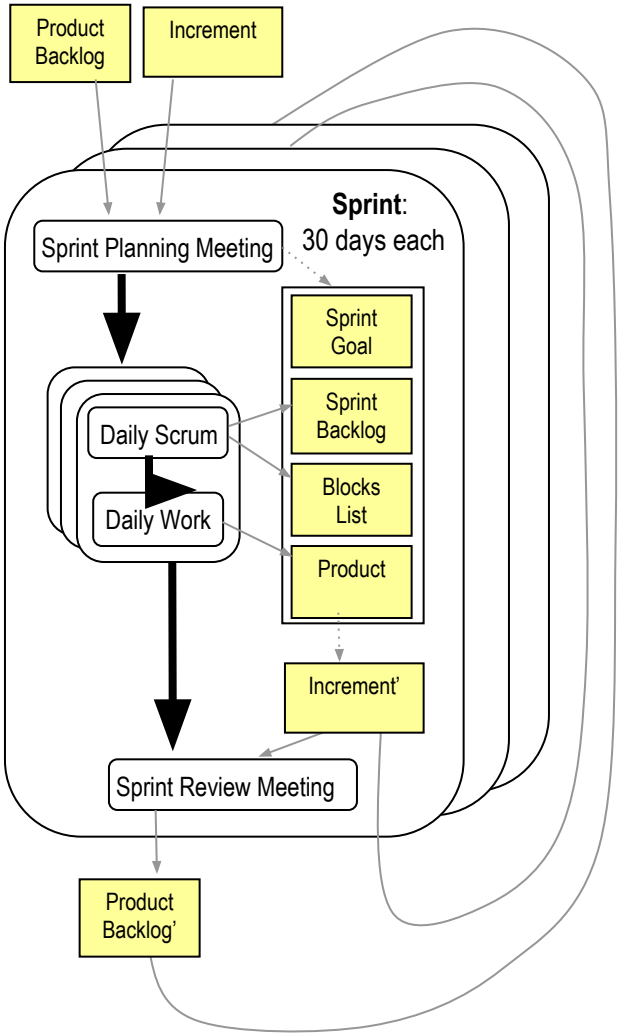
Blocks List
- List of blocks & unmade decisions
- Owned by ScrumMaster
- Updated daily

Increment
- Version of the product
- Shippable functionality (tested, documented, etc.)

Key Meetings

Sprint Planning Meeting
- Hosted by ScrumMaster; ½-1 day
- **In:** Product Backlog, existing product, business & technology conditions
  1. Select highest priority items in Product Backlog; declare Sprint Goal
  2. Team turns selected items into Sprint Backlog
- **Out:** Sprint Goal, Sprint Backlog

Daily Scrum
- Hosted by ScrumMaster
- Attended by all
- Same time every day
- **Answer:** 1) What did you do yesterday? 2) What will you do today? 3) What's in your way?
- Team updates Sprint Backlog; ScrumMaster updates Blocks List

Sprint Review Meeting
- Hosted by ScrumMaster
- Attended by all
- Informal, 4-hour, informational
- Team demos **Increment**
- All discuss
- Hold **retrospective**
- Announce next Sprint Planning Meeting

Development Process